

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B.Tech. Branch : E&CE / E&CSE Semester: III Subject Code & Name: BTESC305_EC Digital Electronics and Microprocessor Max Marks: 60 Date: 09/07/2024 Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	i. Perform the Subtraction by using 1's complement method $(1110101)_2 - (1111)_2$ ii. Perform the Subtraction by using 2's complement method $(1101101)_2 - (1010)_2$	Evaluate	6
B)	Explain and prove De Morgan's theorem in Boolean algebra.	Understanding	6
C)	Conversion of Number System. i) Convert 29.37 Decimal to Binary ii) Convert 45D2.56 Hexadecimal to Octal iii) Convert 6327.405 Octal to Decimal	Evaluate	6
Q.2	Solve Any Two of the following.		12
A)	Write short note on. i. Parity Generator ii. Don't Care Conditions	Applying	6
B)	Compare Multiplexer and Demultiplexer with neat diagram.	Analyze	6
C)	Minimize and implement by using Universal gate using K-Maps. $F(A, B, C, D) = \sum m(1, 2, 6, 7, 8, 10, 13, 14, 15) + d(0, 3, 5, 12)$	Applying	6
Q. 3	Solve Any Two of the following.		12
A)	Design Decade ripple counter.	Create	6
B)	Convert S-R Flip-Flop to J-K Flip-Flop.	Applying	6
C)	What is shift register? Explain in detail any two applications of shift register.	Understand	6
Q.4	Solve Any Two of the following.		12

A)	State and explain with examples the different addressing modes of 8085 microprocessor.	Analyze	6
B)	What is stack pointer and what is its function?	Understand	6
C)	Explain Data Transfer Instructions in 8085 microprocessor.	Analyze	6
Q. 5	Solve Any Two of the following.		12
A)	Write a short note on Procedure and Subroutine.	Understand	6
B)	What is assembly language? Explain assembler, compiler and interpreter.	Understand	6
C)	Explain with example different types of instruction set of 8086	Analyze	6
	*** End ***		